

① najprv určime char polynom $P(\lambda) = \det(A - \lambda I)$

$$\det \begin{pmatrix} -1-\lambda & -2 & 0 \\ -1 & 3-\lambda & -1 \\ 2 & 3 & 1-\lambda \end{pmatrix} = 0 \dots -(\lambda+1)(\lambda-2)^2 = 0$$

$$\lambda_1 = -1 \quad \lambda_2 = 2$$

hľadáme najprv všetky príslušné vektory $(A - \lambda I)v = 0$

pre $\lambda = -1$ $\begin{pmatrix} 0 & -2 & 0 \\ -1 & 4 & -1 \\ 2 & 3 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$ $y=0$ volba $v_1 = (-1, 0, 1)$
 $x = -z$

pre $\lambda = 2$ $(A - 2I)v = 0$ $\begin{pmatrix} -3 & -2 & 0 \\ -1 & 1 & -1 \\ 2 & 3 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$ $v_2 = (2, -3, -5)$

zoberieme vektor pre $\lambda_2 = 2$ $\begin{pmatrix} -3 & -2 & 0 \\ -1 & 1 & -1 \\ 2 & 3 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ -3 \\ -5 \end{pmatrix}$ $w = (0, -1, 2)$

podpriestor U_1 : Báze: $\{(1, 0, 1), (2, -3, -5)\}$ U_2 $B = \{v_2, w\}$ (A. v_2 w mim)

$$\begin{pmatrix} -1 & -2 & 0 \\ -1 & 3 & -1 \\ 2 & 3 & 1 \end{pmatrix} \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \Rightarrow -1v_1 + 0v_2$$

$$A \cdot w = \begin{pmatrix} -1 & -2 & 0 \\ -1 & 3 & -1 \\ 2 & 3 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ -5 \\ -1 \end{pmatrix}$$

$$a_1 \begin{pmatrix} 2 \\ -3 \\ -5 \end{pmatrix} + a_2 \begin{pmatrix} 0 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ -5 \\ -1 \end{pmatrix}$$

$$a_1 = 1 \quad a_2 = 2$$

$$\begin{bmatrix} \mathcal{F} & | & \\ A & v_2 & B & v_2 \end{bmatrix} = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$$

$$\begin{pmatrix} -1 & -2 & 0 \\ -1 & 3 & -1 \\ 2 & 3 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ -3 \\ -5 \end{pmatrix} = \begin{pmatrix} 4 \\ -1 \\ -10 \end{pmatrix} \Rightarrow +0v_1 + 2v_2$$

$$[\mathcal{F}_{A|U_1}]_{B U_1} = \begin{pmatrix} -1 & 0 \\ 0 & 2 \end{pmatrix}$$